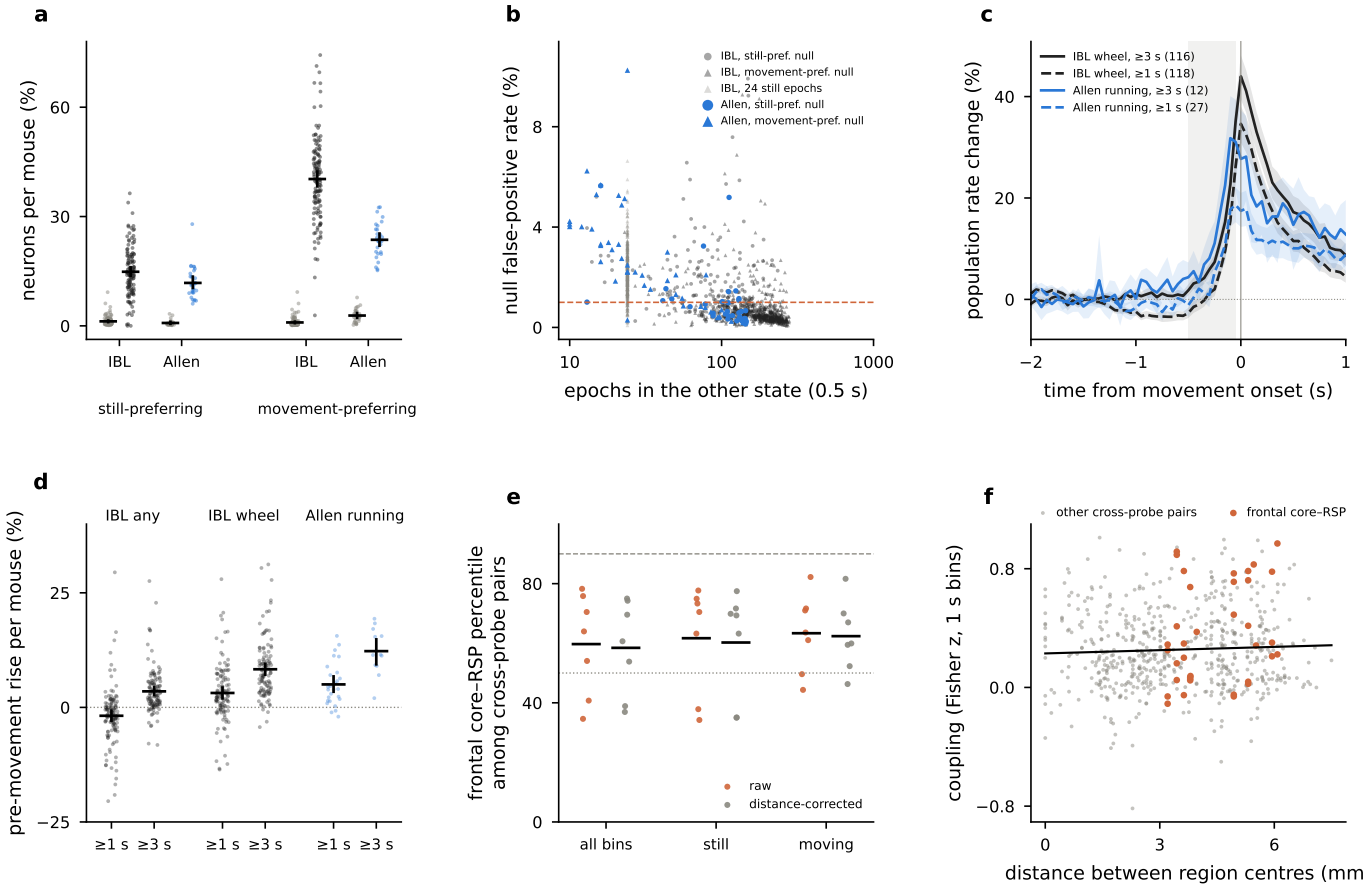


Figure 19: Allen follow-up analyses and their IBL counterparts: label calibration, onset rule and frontal–retrosplenial coupling. **a** Still- and movement-preferring neurons per mouse (colour) against their circular-shift null (grey): all neurons' grey-screen 50 ms counts rotated together against the fixed behavioural labels (200 offsets $\geq$10 s per session) and the epoch test repeated. IBL: 386 sessions, 129 mice, report definition; Allen: Visual Behavior, 44 sessions, 30 mice (follow-up export). Black: mean and 95% bootstrap CI over mice. **b** Null false-positive rate per session against the number of epochs in the opposite state. Grey triangles: IBL movement-preferring null with still epochs subsampled to 24, the Allen median. Dashed: nominal 1%. **c** Population rate around movement onsets preceded by $\geq$3 s (solid) or $\geq$1 s (dashed) of awake stillness, percent change from the -2 to -1 s mean; mean over mice (number in brackets) and 95% bootstrap CI. IBL: wheel onsets (wheel-only states); Allen: running onsets. Shaded: -0.5 to -0.05 s. **d** Rise (-0.5 to -0.05 s against -2 to -1 s) per mouse for each onset rule: IBL any-movement onsets (report definition), IBL wheel onsets and Allen running onsets; sessions with $\geq$3 onsets. **e** Allen OpenScope (8 sessions, 7 mice with a frontal core and an RSP region on different probes): percentile of each mouse's frontal core–RSP pairs among all cross-probe region pairs of the session, from Fisher z of 1 s regional rates (60 s running mean removed), in all, still and moving bins (passive viewing; running $>$2 cm/s; still = no running within $\pm$1 s); raw (orange) and after removing a per-session linear fit on distance (grey). Bars: mean over mice. Dotted: 50th, dashed: 90th percentile. **f** Coupling against the distance between Allen CCF region centres for all 614 cross-probe pairs of the 11 sessions; orange, frontal core–RSP pairs; line, linear fit.